

Fraction of Undetectable Errors

FUE

FUE as small as possible

$$= \frac{\text{total \# undetectable errors}}{\text{total \# possible valid errors}}$$

FUE (M=2)

$$= \frac{\text{total \# undetectable 2-bit errors}}{\text{total \# possible 2-bit errors}}$$

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Error Vector

B

⊕ Suppose we transmit a codeword that has n bits $B = [_ _ _ _ _]_{1 \times n}$

⊕ Define the error vector $\underline{e} = [e_{n-1}, \dots, e_1, e_0]_{1 \times n}$ where:

- ➡ $e_i = 1$ if error occurs to the i^{th} bit position
- ➡ $e_i = 0$ otherwise

received
n-bit block

$$R = B + e$$

↑
"XOR"

$$\begin{array}{l} \downarrow \downarrow \\ \left\{ \begin{array}{l} 1+1=0 \\ 0+1=1 \\ 1+0=1 \\ 0+0=0 \end{array} \right\} \begin{array}{l} \text{error} \\ \text{no error} \end{array} \end{array}$$

⊕ Fraction of Undetectable Errors (FUE)

- ➡ FUE = total # undetectable errors / total # valid errors

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$$k=4, \quad n-k=1, \quad n=5$$

Pattern: even # 1's in the codeword

E.g.

$B = [1 \ 0 \ 1 \ 1 \ 1]_{1 \times 5}$	$R = B + e$	
$R = [0 \ 0 \ 1 \ 1 \ 1]$	$e = [1 \ 0 \ 0 \ 0 \ 0]$	$M=1$: detectable
$R = [1 \ 0 \ 0 \ 0 \ 1]$	$e = [0 \ 0 \ 1 \ 1 \ 0]$	$M=2$: undetectable
$R = [0 \ 0 \ 0 \ 1 \ 0]$	$e = [1 \ 0 \ 1 \ 0 \ 1]$	$M=3$: detectable
$R = [0 \ 1 \ 0 \ 0 \ 1]$	$e = [1 \ 1 \ 1 \ 1 \ 0]$	$M=4$: undetectable
$R = [0 \ 1 \ 0 \ 0 \ 0]$	$e = [1 \ 1 \ 1 \ 1 \ 1]$	$M=5$: detectable
	$e = [0 \ 0 \ 0 \ 0 \ 0]$	$M=0$: no error

* undetectable errors: $M=2, M=4$
 "even # error positions"

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$$\begin{aligned}
 FUE &= \frac{\text{total \# undetectable errors}}{\text{total \# valid errors}} & e &= [_ _ _ _ _]_{1 \times 5} \\
 &= \frac{\binom{5}{2} + \binom{5}{4}}{\text{total \# non-zero error vectors}} = \frac{10 + 5}{2^5 - 1} \\
 &= \frac{10 + 5}{31} = \frac{15}{31} \\
 FUE (M=2) &= \frac{\binom{5}{2}}{\binom{5}{2}} = 100\%
 \end{aligned}$$

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5. (30 points) Suppose that two check bits are added to four information bits (i_3, i_2, i_1, i_0). The first check bit c_1 is the even parity check of the first two information bits (i_3, i_2), and the second check bit c_0 is the even parity check of the final three information bits (i_2, i_1, i_0). The codeword is ($i_3, i_2, i_1, i_0, c_1, c_0$).
- a. (15 points) What fraction of errors is undetectable? Justify your answer. $FuE = ?$
- b. (15 points) What fraction of 2-bit errors is undetectable? Justify your answer. $FuE(M=2) = ?$

$$k=4 \quad n-k=2 \quad n=6$$

$$B = [\underbrace{i_3, i_2}_{\text{check } c_1}, \underbrace{i_1, i_0, c_1}_{\text{check } c_0}, c_0]_{1 \times 6}$$

$$c_1 = (i_3 + i_2) \bmod 2$$

$$c_0 = (i_2 + i_1 + i_0) \bmod 2$$

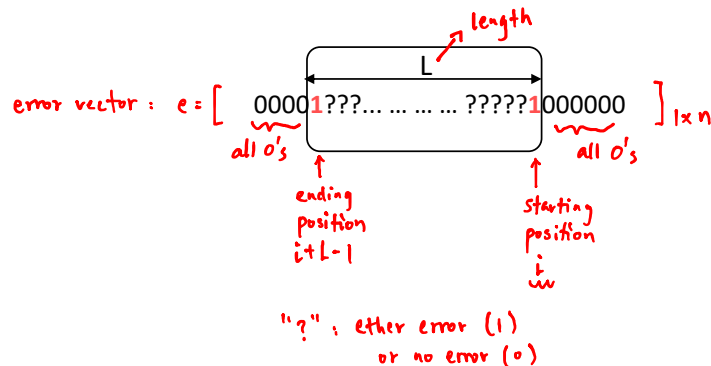
$$e = [_ _ _ _ _ _]_{1 \times 6}$$

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Error Burst

Errors can be classified according to:

- ➡ Number of bit error positions: M-bit error
- ➡ Separation of bit error positions: error burst of length L
 - Error starts at bit position i and ends at bit position $(i + L - 1)$



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$$e = [1 \ 0 \ 0 \ 0 \ 0]_{1 \times 5} \quad L=1 \equiv M=1$$

$$e = [0 \ 0 \ 1 \ 1 \ 0] \quad L=2 \ ? \ M=2$$

$$e = [1 \ 0 \ 0 \ 0 \ 1] \quad L=5 \quad M=2$$

$$e = [1 \ 0 \ 1 \ 0 \ 0] \quad L=3 \quad M=2$$

$$e = [1 \ 0 \ 1 \ 1 \ 1] \quad L=5 \quad M=4$$

$\uparrow \qquad \qquad \qquad \uparrow$

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$$FUE(L) = \frac{\text{total \# undetectable error bursts of length } L}{\text{total \# error bursts of length } L} = \frac{(n-L+1) \times 2^{L-2}}{\text{diff}}$$

Starting position $i : 0, 1, \dots, \underline{n-L} : \underline{(n-L+1)} \text{ options}$

$$e = [\underbrace{0 \ 0 \ 0 \ \dots \ 0}_L \underbrace{?? \ \dots \ ?}_L \underbrace{1 \ 0 \ 0 \ \dots \ 0}_L]_{1 \times n}$$

$\uparrow \qquad \qquad \qquad \uparrow$
 $i+L-1 \qquad \qquad \qquad i$
 $i+L-1 \qquad \qquad \qquad i = \underline{n-L}$
 $= n-1$

"?" terms : $L-2 \Rightarrow \underline{2^{L-2}}$

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